

Challenges related to Ducts, Chimneys and Tail-end Alignment

Internal Inspection of Exhaust Gas Chimney during running condition

Many lined & unlined steel chimneys, in manufacturing industries, are inspected periodically to assess structural integrity by measuring the shell plate wall thickness. These chimneys measure from 1.2 to 9 meters in diameter with height from 10 meters to 86 meters and wall thickness varies from 6mm to 32mm. Internal and external inspection needs to be carried out to assess the condition of the chimney. Currently the inspection is carried out manually & rope access method with a lot of care (PPE) and with a proper shutdown. Inside inspection of chimney poses a difficult challenge as the person needs to enter the confined space for inspection. Visual inspection was tried out in the past by lowering a camera with light arrangement (using ropes) in running condition. However, the quality of images was poor and therefore couldn't assess the condition of the chimney.

We are thus seeking a solution to inspect (assess the condition and measure wall thickness) the chimney in running condition as it saves time and avoids manual intervention completely.